

Teo Layer

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I am an advanced Systems Engineering student with over a year of hands-on independent project experience, driven by a genuine passion for building functional, scalable, and well-structured software. My academic background spans key disciplines including Systems Analysis, where I design UML diagrams—use cases, class diagrams, and sequence diagrams—derived from functional and non-functional requirements, and Systems Design, where I apply design patterns such as MVC, Repository, Observer, and Singleton to architect robust, maintainable solutions following software engineering best practices. These foundations provide me with a strong theoretical grounding that I continuously reinforce through real-world, self-directed development work.

Over the past year, I independently designed and built Dental Care Hub, a full-featured dental practice management system developed with React 18, TypeScript, and Vite. The application covers patient management, appointment scheduling, financial transaction tracking, inventory control with low-stock alerts, and an interactive dental odontogram component. I structured the project using a layered architecture: the Context API for global state management across six distinct domain providers, a dedicated service layer isolating business logic, and Axios for HTTP communication with JWT-based authentication interceptors. The user interface was built with Tailwind CSS, Radix UI, and shadcn/ui component libraries, while React Hook Form paired with Zod schema validation ensured robust and type-safe form handling. Data visualization dashboards were implemented using Recharts, delivering real-time KPIs to end users.

Complementing my frontend expertise, my academic training also includes Python and C, languages through which I have developed strong algorithmic thinking, memory management fundamentals, pointer arithmetic, and efficient data processing capabilities. I also hold solid knowledge in relational database design, covering entity-relationship modeling, SQL query writing, normalization techniques, and database integration patterns within layered application architectures.

I am a self-motivated professional fully comfortable working autonomously in remote environments. I adapt quickly to new technologies, take complete ownership of development cycles from requirements gathering and system modeling to implementation, testing, and delivery, and communicate clearly and effectively across distributed teams. I am genuinely eager to contribute my technical skills, analytical mindset, and software engineering knowledge to a dynamic remote role where I can keep growing as a developer.